Thus the number of ways in which the 4 places can be filled, by the rule of product is $4 \times 3 \times 2 \times 1 = 24$. Hence, the required number of strings is 24.

(ii) There are 5 different letters with which 5 places are to be filled. The first place can be filled in 5 ways as any one of the five letters P,R,I,M,E can be placed there. Having filled the first place with any of the 5 letters, 4 letters are left to be placed in the second place, three letters are left for the third place and 2 letters are left to be put in the fourth place. The remaining 1 letter has to be placed in the fifth place.

Hence, the total number of ways filling up five places is $5 \times 4 \times 3 \times 2 \times 1 = 120$.



Observe the similarity between the above two cases.

Example 4.9 How many strings of length 6 can be formed using letters of the word FLOWER if

(i) either starts with F or ends with R?

(ii) neither starts with F nor ends with R?

Solution:

In any such string, each of the letters F,L,O,W,E,R is used exactly once.

(i) If such a string starts with F, then the other five positions are to be filled with the letters L, O, W, E, R.

As there cannot be any repetition of letters in the formation of the strings we can fill up the 2nd, 3rd, 4th, 5th and 6th places in 5, 4, 3, 2 and 1 ways.

Hence, by the rule of product, the number of strings of length 6 starting with F is equal to $5 \times 4 \times 3 \times 2 \times 1 = 120$.

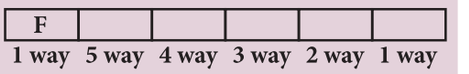


Figure 4.4

If such a string ends with R, then the other five positions are to be filled with the letters F,L,O,W,E.

As in the previous case, we conclude that the number of strings of length ending with R is 120.

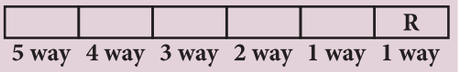


Figure 4.5

If a string starts with F and also ends with R, then the other 4 positions are to be filled with letters L, O, W, E.

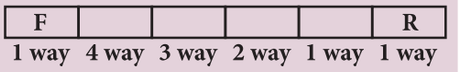


Figure 4.6

As in the previous cases, the number of strings of length of 6 starting with F and ending with R is $4 \times 3 \times 2 \times 1 = 24$.

By the principle of inclusion - exclusion, the number of strings of length 6, either starting with F or ending with R is $120 + 120 - 24 = 216$.

(ii) A string that neither starts with F nor ends with R is one which has not been counted in (i). Together, they account for all possible strings of length 6 formed out of the letters, F,L,O,W,E,R, where no letter is repeated.

Now, the number of all such strings is formed by filling the first position by any of the 6 letters, the second by any of the remaining 5 letters and so on. That is, there are in total $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ such strings. The number of words neither starting with F nor ending with R is the same as the difference between total number of letter strings and the number of strings either starting with F or end with R which is $720 - 216 = 504$.